

Reliability audit

Android performance, indexing, sync, storage, and current controls

COMPLETE · 31 / 31 TASKS

54 s

first 2,001-entry index

440–894 ms

repeat cold display

0

idle scans in 303 s

0

dropped dashboard frames

Executive result

The reported “thinking” loop was reproducible as a real Android startup race. Selecting the first folder opened the vault twice; the second open killed a worker while SAF replies were still in flight, causing a native Flutter did_send abort. The duplicate open is removed, worker shutdown now drains cooperatively, and the same guarded profile fixture stayed usable through startup, rebuild, idle, storage contention, and control tests.

What was found

Issue	Observed behavior and cause	Disposition
Startup crash	First Android folder selection opened the vault twice; the second worker replacement killed a live SAF response port.	FIXED
Worker teardown	Disposal could strand a command or kill an isolate during native/SAF I/O.	FIXED
Repeated indexing	A settled idle vault could be rescanned by overlapping lifecycle/sync triggers.	FIXED
UI stalls	Index serialization, validation, search construction, and storage work competed with the root isolate.	REDUCED
SAF tail latency	Large attachment writes overlap interactive reads and atomic saves.	MONITORED
Dashboard polling	Unchanged trace reloads consume SAF reads while the dashboard is open.	MEASURED

Device evidence

The guarded profile build ran on a Nothing A024 (Android 16 / SDK 36) using the disposable TyLogAuditVault tree. The production vault was never selected or modified during measurements.

Measurement	p50	p95	max / result
SAF note read + atomic save / idle	263 ms	325 ms	325 ms
SAF with 24 MiB attachment transfer	264 ms	545 ms	744 ms
Dashboard load + parse	46.5 ms	56.5 ms	60.9 ms
Dashboard JSON parse only	1.19 ms	2.12 ms	5.84 ms
Inline vs worker root-isolate stalls	6 frames	3 frames	63 vs 35 ms

Idle behavior

After settling, index.json and search-index.json.gz stayed unchanged for 303 seconds.

Integrity

All 60 SAF mutations preserved bytes, hashes, decoded content, and atomic replacement semantics.

Implemented changes

1. First-vault onboarding returns after the selected entry has already been opened, preventing the duplicate startup open.
2. Worker disposal settles the client stream immediately but keeps the worker isolate alive until its current native/storage await returns.
3. Successful search-index publication is explicit and revisioned; cancelled or failed builds do not publish partial search state.
4. Save failures now gate note/vault navigation so dirty buffers are retained.
5. Sync authentication failures pause automatic retries until credentials change or the user explicitly retries.
6. Report exports request only canonical compiler dependencies, avoiding unrelated vault media reads.

Verification coverage

Software

599 Flutter tests passed with one intentional skip; 209 core tests, 19 Rust tests, the 10k-note benchmark, five macOS PDF tests, targeted analysis, and graph refresh passed.

Connected phone

SAF contention, worker shutdown/replacement, background/foreground sync, native worker indexing, PDF export, and all Magic editor actions passed in profile mode.

Operational recommendation
Keep large attachment transfers away from interactive editing windows. The measured p95 SAF tail rose by 68% under contention even though content integrity remained intact. No lock or atomic-write change is justified by the current evidence.

Limits and follow-up

Network baseline	The disposable fixture had no Nextcloud account, so a real remote sync completion time and production 401/403 run were unavailable. Controlled 401/403 regressions verify that automatic retries pause correctly.
Before/after timing	No authentic pre-fix phone baseline exists. This report therefore makes no percentage-improvement claim; it reports post-fix measurements and the reproduced crash cause.
App-local state	An early driver command without <code>--keep-app-running</code> removed app-local settings. The external production vault was untouched, but its registry entry and credentials may need to be restored on the phone.

Evidence: docs/audits/2026-09-06/evidence/ — T00, T25, T26, T28, and T30. Layout guidance retrieved with TypstRAG from the Typst Page Setup Guide, template guidance, semantic PDF guidance, and table/syntax references (Typst v0.15.1).